PROBLEM STATEMENT

1. AI-Powered Drone Navigation for Emergency Medicine Delivery

In disaster-affected regions such as flood areas, delivering emergency medicines quickly and safely is a challenging task due to dynamic environmental conditions. Roads may become inaccessible, weather conditions may change, and movement costs may vary depending on terrain. The drone must make real-time routing decisions using incomplete information while balancing travel cost, safety, and delivery time. Therefore, an efficient AI search algorithm is required to find optimal and reliable paths under uncertainty.

2. Smart City Traffic Signal Optimization

Managing traffic flow in a smart city with hundreds of intersections is a complex optimization problem. The system must continuously adjust traffic signal timings to minimize waiting time, fuel consumption, and congestion. Due to the large search space and continuously changing traffic patterns, traditional search techniques may become inefficient and may get trapped in locally optimal solutions. An intelligent optimization approach is required to generate scalable and adaptive traffic management solutions.

3. Autonomous Mars Rover Navigation

An autonomous rover exploring Mars must navigate unknown terrain while collecting samples and avoiding hazards such as rocks and craters. Since the rover operates with limited sensing capability and incomplete environmental information, it cannot depend on predefined maps. The system requires real-time decision-making, continuous learning, and adaptive search strategies to safely explore uncertain environments and update its knowledge during operation.

4. AI-Based University Exam Timetable Generation

Creating an examination timetable for a university involves handling multiple constraints including student availability, faculty schedules, limited examination halls, subject dependencies, and special accommodations. As the number of courses and students increases, manual scheduling becomes complex and inefficient. An AI-based Constraint Satisfaction Problem (CSP) approach is required to automatically generate valid timetables while satisfying all necessary constraints and improving scheduling efficiency.

5. Strategic Game AI for Real-Time Decision Making

Developing an intelligent game opponent requires AI systems to make optimal decisions against human players within strict time limitations. The game environment contains a large number of possible moves, resulting in extremely large decision trees that increase computational complexity. An efficient decision-making algorithm is required to evaluate game states, reduce unnecessary computation, and balance optimal strategies with real-time performance requirements.